
Girl with a Pearl Earring via Genetic Algorithms

Project Report



GROUP C35 - Convergence

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1 Introduction

This report describes the design, implementation and experimental evaluation of a Genetic Algorithm (GA) for reconstructing Vermeer’s *Girl with a Pearl Earring* using exactly 100 semi-transparent triangles. The target is a 300×400 pixel image and the objective is to minimise the pixel-level Root Mean Squared Error between the rendered candidate and the original painting.

2 System Design and Implementation

The codebase was organised into separate `.py` modules following object-oriented principles, making it easy to swap operators or add new strategies without modifying the core engine.

2.1 Representation

2.1.1 The Gene: *Triangle*

Each gene is a `Triangle`, defined by three floating-point (x, y) vertices and an RGBA color $(R, G, B, A) \in [0, 255]^4$. Vertex coordinates are stored as floats throughout evolution and only rounded at render time, preserving the continuous search space. The alpha channel was included to allow blending between layers. Without it, smooth gradients and soft transitions would be impossible to approximate. Finally, `Triangle` objects are immutable: all operators produce new instances rather than modifying existing ones and triangles with area below 1 px^2 are flagged as degenerate and can be penalised during mutation.

2.1.2 The Chromosome: *Individual*

An `Individual` is an ordered list of exactly 100 `Triangle` objects. Draw order is part of the genotype: index 0 is the bottom layer, index 99 the top. Unlike `Triangle`, which enforces immutability at the language level, `Individual` achieves it through encapsulation, ensuring no individual is ever in an intermediate state during the generational loop. Fitness is computed once on first access and cached, avoiding redundant renders across selection, sorting and logging.

2.1.3 Initialisation Strategies

Six strategies were implemented, ranging from fully random to spatially structured (A.1).

Table 1: Initialisation strategies

Strategy	Key property
<code>random</code>	Uniform random geometry and color; maximum diversity
<code>random_semitransparent</code>	Restricts $\alpha \in [30, 120]$, encouraging layering of triangles from different opacities from the beginning
<code>random_small</code>	Generates small triangles by keeping vertices within 15% of the canvas size from each other
<code>from_grid_random_color</code>	10×10 grid with Gaussian vertex noise; full canvas coverage
<code>random_sorted_alpha</code>	Opaque triangles placed on lower layers
<code>random_quadrant</code>	Vertices constrained per 5×5 cell; guarantees per-region coverage

In grid-based strategies, Gaussian vertex noise is applied per individual to avoid all chromosomes sharing identical positions from generation 0.

2.2 Fitness Function

The fitness function follows the Strategy pattern: an abstract base class defines a single `evaluate(candidate) -> float` interface, making the metric swappable without modifying any other component. The primary metric is **RMSE**:

$$\text{RMSE} = \sqrt{\frac{1}{H \times W \times C} \sum_{h,w,c} (I_{\text{rendered}}[h, w, c] - I_{\text{target}}[h, w, c])^2}$$

The target array is pre-cast to `float32` once to avoid repeated conversions. A perfect reconstruction scores 0; the theoretical maximum is 255.

2.3 Rendering Pipeline

The rendering pipeline, implemented in `gautils.py`, converts a chromosome into a pixel array that can be compared against the target. The core function `render()` composites 100 triangles in draw order onto a 300×400 black canvas. Since PIL's `ImageDraw.polygon()` does not support per-polygon alpha blending, each triangle is drawn onto a separate transparent RGBA layer and composited onto the canvas via the Porter-Duff *over* operator [1], which is the standard algorithm for alpha compositing in computer graphics. A black background was chosen as a neutral baseline: uncovered regions contribute zero signal to the fitness evaluation, avoiding any artificial penalty for areas the triangles have not yet reached (A.2).

2.4 Genetic Operators

2.4.1 Selection

Two strategies were implemented (A.3.1). Tournament selection was adopted as the primary strategy because it requires no global fitness scaling and selection pressure is directly controlled by a single parameter k [2](A.3.1).

Table 3: Selection strategies.

Strategy	Description
<code>TournamentSelection</code>	Samples k individuals and selects the best; pressure controlled by k
<code>RankSelection</code>	Assigns probabilities by fitness rank rather than raw value; avoids dominance of very fit individuals

2.4.2 Crossover

Five crossover operators are implemented, all exchanging complete `Triangle` objects - a triangle is never split across offspring, as combining unrelated vertex and color data would disrupt building blocks. Crossover is applied with probability $p_c = 0.8$ [2]; when it does not apply, parents are copied directly to offspring before mutation (A.3.2).

Uniform crossover was selected as the baseline because every triangle interacts with all others

Table 5: Crossover operators.

Operator	Description
SinglePoint	One random cut point; everything after it is exchanged between parents
Uniform(p)	Each triangle independently taken from one parent with probability p
KPoint(k)	k cut points; preserves smaller blocks while allowing more recombination
SegmentShuffle	Random number of cuts; each segment independently assigned to a child
Blend(α)	Interpolates parents' genes to create new triangles and extends by α [3]; can produce values outside parent range

through alpha compositing, making the *position problem* [2] especially relevant.

2.4.3 Mutation

Five mutation operators were implemented, with Gaussian mutation as the baseline [2]. Mutation is applied per triangle at rate 0.05 (A.3.3).

Table 7: Mutation operators.

Operator	Description
Gaussian(σ)	Perturbs vertices and colors drawn from $\mathcal{N}(0, \sigma^2)$; can make large jumps, good for local search
Creep(δ)	Similar to Gaussian but noise is capped at $\pm\delta$; large jumps are impossible, better for fine-tuning
Reset	Replaces selected triangles with new random ones; injects diversity
Swap	Exchanges two triangles in the draw order without altering geometry or color
Composite	Chains multiple operators in sequence

Three decay schedulers anneal the noise parameters exponentially over the run (SigmaDecay for Gaussian, DeltaDecay for Creep, CompositeSigmaDecay for the Gaussian inside a Composite). This produces broad exploration early and fine refinement later.

2.5 Population Management

The population is set with elitism preserving the best n_{elites} unchanged each generation. Four diversity metrics are tracked throughout the run: phenotypic variance (spread of fitness values), genotypic variance (spread of chromosome distances to the best individual) and their respective entropies [2]. These feed into a DiversityAwareEarlyStopping mechanism that only terminates a run when fitness stagnation and diversity collapse are observed simultaneously, preventing premature termination of runs where the population is still genetically diverse enough to produce improvements.

2.6 GA Engine

The GA runs a standard generational loop: select parents, apply crossover and mutation to produce offspring, keep the best individuals via elitism and replace the rest of the population. After each generation, statistics are logged and early stopping is checked. The decay schedulers update their noise parameters through a callback at the end of each generation. Because all operators follow the same abstract interface, any strategy can be swapped without changing the engine itself.

3 Experimental Methodology

3.1 Protocol Overview

Experiments follow a sequential **OFAT** approach [4] across five initial phases, each varying a single component while holding all others fixed. Since OFAT cannot detect factor interactions, seven additional phases re-test earlier decisions with improved operators and explore promising combinations. Most configurations were run with three seeds; some later phases were limited to fewer due to computational cost.

3.2 Phase Structure

Phase 1 - Baseline: The baseline configuration represents the most neutral choices in the GA literature and serves as the reference point for all subsequent phases:

Table 9: Baseline configuration (Phase 1).

Component	Choice	Rationale
Selection	TournamentSelection ($k=3$)	Moderate pressure, no fitness scaling
Crossover	UniformCrossover	No positional assumptions, maximum mixing
Mutation	GaussianMutation (fixed σ)	Natural for continuous spaces
Elitism	5	Retains top solutions without sacrificing diversity.
Population	50	Balance between diversity and compute budget

Phase 2 - Elitism sweep ($k \in \{0, 1, 3, 5\}$): Quantifies the effect of elitism on convergence speed and final fitness. Higher elitism protects against regression but risks reducing population diversity by repeatedly propagating the same genetic material.

Phase 3 - Selection sweep: Tests TournamentSelection with $k \in \{2, 5, 10\}$ and RankSelection ($sp=1.5$), holding all other components at their Phase 1 values.

Phase 4 - Crossover sweep: Tests all five crossover operators. The key question is whether locality-preserving operators (SinglePointCrossover, KPointCrossover) outperform UniformCrossover for this problem, where draw order is semantically meaningful.

Phase 5 - Mutation sweep: Tests Gaussian mutation with fixed vs. decaying σ , creep mutation, composite mutation (Gaussian + Reset + Swap) and the full adaptive composite with σ decay.

Phases 6-12 extend the study by re-testing earlier decisions:

Table 10: Extended phases beyond OFAT.

Phase	Description
6	Best configuration with 20 000 generations (convergence ceiling)
7	Initialisation strategy sweep (6 strategies, best operators)
8	Elitism re-test with best operators from Phases 3-5
9	Interaction check: Tournament ($k=5$) with best mutations
10	Grid search over $\text{mutation_rate} \times \text{vertex_sigma_max}$
11	Population size sweep ($N \in \{100, 150\}$) with best configuration
12	Best configuration with $N \in \{150, 300\}$ and 20 000 generations

4 Results and Discussion

4.1 Phase-by-Phase Results

Phase 1 - Baseline

The baseline configuration (TournamentSelection $k=3$, UniformCrossover, fixed Gaussian mutation, 5 elites) achieves a mean RMSE of 29.69 ($\sigma=0.75$) across three seeds, with a best seed of 29.02. RMSE falls rapidly from above 70 to about 35 in the first 500 generations, then slows and plateaus near generation 3 000. The seeds follow similar trajectories, suggesting stable behaviour and limited seed noise (B.1).

Phase 2 - Elitism Sweep

Testing $n_elites \in \{0, 1, 3, 5\}$ reveals a monotone relationship: more elites consistently yields better final fitness. $n_elites=5$ achieves the best mean RMSE (29.69, $\sigma=0.75$), while no-elitism ($n_elites=0$) is the clear outlier at 32.72 ($\sigma=0.38$), confirming that preserving at least one elite per generation is beneficial. $n_elites=5$ is carried forward, with a re-test planned for Phase 8 once better operators are established.

Phase 3 - Selection Sweep

This phase gives the clearest OFAT separation. `tournament_k10` performs best (27.84 RMSE, $\sigma = 0.62$), followed by `tournament_k5` (28.35, $\sigma = 0.54$), while `tournament_k2` and `RankSelection` lag behind at 30.50 and 32.89. The gap suggests that stronger selection pressure is beneficial here: with $N = 50$, $k = 10$ samples 20% of the population per selection event. Its low per-seed variation supports carrying `tournament_k10` forward.

Phase 4 - Crossover Sweep

All five crossover operators converge to nearly identical final RMSE values, with means ranging from 27.73 (`BlendCrossover`) to 27.97 (`SinglePointCrossover`) - a spread of under 0.25 RMSE across the full phase. The convergence curves are indistinguishable after generation 1 000. `BlendCrossover` is carried forward for its lowest mean and tightest variance ($\sigma=0.30$).

Phase 5 - Mutation Sweep

Phase 5 shows the largest OFAT spread, with over 3 RMSE points between the best and worst configurations. `gaussian_decay` achieves the best mean RMSE (25.05, $\sigma=0.70$), suggesting that decay is preferable to fixed sigma: both `gaussian_decay` (25.05) and `creep_decay` (26.18) outperform `gaussian_fixed` (27.90) and `creep_fixed` (28.19) by roughly 2–3 RMSE points. This supports the benefit of annealing, combining early exploration with late-stage refinement. Composite operators add no clear benefit: `composite_gaussian_decay` (25.79) is competitive but likely suffers from the disruption introduced by `ResetMutation` and `SwapMutation` at $N=50$. Therefore, `gaussian_decay` is carried forward, improving by approximately 2.8 RMSE over the baseline (B.2).

Table 12: Summary of OFAT phase winners and mean RMSE across 3 seeds.

Phase	Winner	Mean RMSE	Std
1 - Baseline	tournament_k3, uniform XO, gaussian fixed, 5 elites	29.69	0.75
2 - Elitism	n_elites=5	29.69	0.75
3 - Selection	tournament_k10	27.84	0.62
4 - Crossover	BlendCrossover	27.73	0.30
5 - Mutation	gaussian_decay	25.05	0.70

Phase 6 - Extended Budget

Extending the best OFAT configuration to 20 000 generations drops the mean RMSE from 25.05 to 22.25 ($\sigma=0.63$), corresponding to a reduction of ≈ 2.8 RMSE points under identical operators. The convergence curve shows no plateau at generation 3 000; diversity oscillates in bursts through the first 10 000 generations before collapsing near generation 18 000, confirming that the 3 000-generation budget in Phases 1–5 was a binding constraint.

Phase 7 - Initialisation Strategy Sweep

Six initialisation strategies are evaluated over 500 generations. `quadrant` (mean RMSE 30.25, $\sigma=0.71$) and `semi_transparent` (30.25, $\sigma=0.47$) perform best, while `small_random` (32.26) underperforms. Differences shrink within the first 100 generations under `tournament_k10`, suggesting relative robustness to initialisation. A longer single-seed run of `quadrant` reaches RMSE 24.46 after 3 000 generations, comparable to the best Phase 5 result. `quadrant` is therefore carried forward for its structured canvas-wide coverage from generation 0.

Phase 8 - Elitism Re-test

Re-testing $n_elites \in \{1, 3, 7, 10\}$ with the full best operator set changed the trend observed in Phase 2. In this setting, lower elite counts performed better, with `n_elites=3` obtaining the best observed fitness value (24.35). A plausible explanation is the interaction with `tournament_k10`. Since each tournament samples 20% of the population per selection event, selection pressure is already high. Under these conditions, a larger elite pool may preserve too many similar high-ranking individuals, reducing the effective offspring pool and limiting exploration. However, because each configuration was evaluated using a single seed, this explanation should be treated as an interpretation rather than a confirmed causal effect. Based on the best observed result in this phase, `n_elites=3` is carried forward.

Phase 9 - Interaction Check: Selection $\times$ Mutation

Testing `tournament_k5` paired with `gaussian_decay` (mean RMSE 27.03) and `creep_decay` (28.44) confirms that neither variant outperforms the Phase 3 winner. `tournament_k10 + gaussian_decay` is confirmed as the selection–mutation pair.

Phase 10 - Grid Search: `mutation_rate` $\times$ `vertex_sigma_max`

The dominant factor in the 4×3 grid is mutation rate. `rate=0.01` occupies the three best positions, with the best result at `sigma_max=80` (RMSE 22.99), while `rate=0.2` gives the three worst results (≈ 31 – 32 RMSE), a gap of roughly 9 points. Within `rate=0.01`, larger

sigma performs better, since only about one triangle per individual is mutated per generation, requiring larger perturbations to preserve exploration. In contrast, `rate=0.2` mutates about 20 triangles per generation, overwhelming selection regardless of step size. Therefore, `rate=0.01`, `sigma_max=80` is carried forward.

Phase 11 - Population Size Sweep

Increasing the population size improves performance, with `pop=150` outperforming `pop=100` by about 0.55 RMSE (21.73 vs. 22.28) and showing lower variance ($\sigma=0.09$ vs. 0.34). Its diversity remains active until around generation 1 500, with two recovery bursts, whereas earlier phases collapsed before generation 500. This suggests that a larger population preserves diversity for longer and reduces sensitivity to initialisation. `pop=150` is carried forward.

Phase 12 - Final Configuration

The best configuration (`tournament_k10`, `BlendCrossover`, `gaussian_decay`, 3 elites, quadrant initialisation, `mutation_rate=0.01`, `sigma_max=80`) is run at 20 000 generations with populations of 150 and 300. `pop=300` achieves RMSE 17.93, terminating early at generation 9 937; `pop=150` reaches 19.45 at $\approx 9\,300$ generations. The convergence curve is continuous with no clear plateau. The sigma decay window was deliberately decoupled from the generation budget (`decay_n_generations=5000`), ensuring the transition from exploration to exploitation occurs at an appropriate moment rather than prematurely. The final result represents a 39.6% improvement over the baseline RMSE of 29.69. Visually, the reconstruction captures the painting's broad structure - dark background, silhouette, blue headscarf and warm clothing tones - while fine details like facial features and the pearl earring remain limited by the 100-triangle budget rather than the optimisation itself (B.3).

Table 13: Final configuration and per-phase best results.

Phase	Configuration / Winner	Best Mean RMSE
6 - Extended budget	Best OFAT, 20 000 gens, <code>pop=50</code>	22.25
7 - Initialisation	quadrant	24.46
8 - Elitism re-test	<code>n_elites=3</code>	24.35
9 - Interaction check	<code>tournament_k10</code> + <code>gaussian_decay</code>	27.03
10 - Grid search	<code>rate=0.01</code> , <code>sigma_max=80</code>	22.99
11 - Population sweep	<code>pop=150</code>	21.73
12 - Final	<code>pop=300</code> , 20 000 gens	17.93

4.2 Statistical Validation

To validate the improvements found through the OFAT process, we apply formal statistical hypothesis testing, following the methodology recommended in [2]. The comparison was conducted between the best model found in Phase 12 and the Phase 1 baseline. Both configurations were run under identical conditions: 30 independent runs (seeds 0-29), population size 50, 1 000 generations and RMSE as the fitness function. Both generation count and population size were reduced for computational feasibility.

Before choosing a test, we verified whether the 30 RMSE values from each configuration followed a normal distribution using the Shapiro-Wilk test, which is better suited for small samples ($n=30$). Both samples passed the normality check ($p = 0.68$ for the best model, $p = 0.84$ for the baseline),

allowing us to use a parametric test. We then applied the Welch t-test, a variant of Student’s t-test that does not assume equal variances.

Table 14: Descriptive statistics and hypothesis test results (30 runs each).

	Best Model (p12)	Baseline (p1)
Mean RMSE	26.88	32.31
Std RMSE	0.51	0.95
Welch t-test: $t = -27.53$, $p < 0.001$ (***)		Cohen’s $d = 7.11$

The best model achieves a mean RMSE of 26.88, compared to 32.31 for the baseline, a reduction of approximately 16.8%. The standard deviation is roughly half that of the baseline, indicating more consistent convergence across runs. The Welch t-test confirms the difference is highly significant ($p < 0.001$) and the effect size (Cohen’s $d = 7.11$) is well above the threshold considered ‘large’, meaning the two distributions are almost entirely non-overlapping (B.4).

5 Challenge 1: Alternative Fitness Functions

The additional challenge selected is **Challenge 1**: replacing the standard RMSE fitness function with perceptually motivated alternatives and comparing the results.

5.1 Part I - Single-Objective Fitness Comparison

Three independent GAs were trained with the best operator configuration found in previous phases, optimised for RMSE, differing only in fitness function: **RMSE**, **CIEDE2000** [5] and **SSIM** [6].

RMSE treats all pixels as equally important regardless of where they are in the image: a colour error in the dark background is penalised the same as the same error on the face. The GA therefore optimises towards global pixel accuracy without any notion of perceptual salience. CIEDE2000 operates in the perceptually uniform CIELAB colour space and applies scaling factors (S_L , S_C , S_H) that reduce penalties where the eye is less sensitive, naturally redirecting effort towards perceptually salient regions. SSIM measures structural similarity through local windows rather than per-pixel colour; it was included to test whether a structure-oriented signal is viable as a fitness function for this task. Each best individual was cross-evaluated on all three metrics to ensure a fair comparison.

Table 15: Cross-evaluation results — mean $\pm$ std across seeds 42, 43, 44.

Trained with	RMSE $\downarrow$	CIEDE2000 $\downarrow$	SSIM sim. $\uparrow$
RMSE	23.25 $\pm$ 0.24	8.86 $\pm$ 0.40	$\approx$ 0.51
CIEDE2000	29.18 $\pm$ 0.93	7.62 $\pm$ 0.24	$\approx$ 0.51
SSIM	45.14	15.15	0.532

RMSE and CIEDE2000 each win on their own metric and neither dominates globally: the two GAs appear as approximate mirror images across all three axes (21). RMSE achieves lowest pixel error but pays a 16% perceptual colour penalty; CIEDE2000 achieves a 14% improvement on perceptual colour accuracy at the cost of higher pixel error. Visually, RMSE produces finer detail while CIEDE2000 yields more faithful skin tones and headscarf colour (18). The per-region

analysis shows the gap between RMSE and CIEDE2000 widens in Body, Face and Headscarf relative to the Background and Body, suggesting CIEDE2000 accepts higher pixel error in complex regions in favour of perceptual colour fidelity (23).

CIEDE2000 converges with the lowest seed-to-seed variance ($\text{std} = 0.24$ vs 0.40 for RMSE) (19, 17). The evolution snapshots reveal why: CIEDE2000 locks in correct colours by generation 500, resolving colour placement before refining geometry, whereas RMSE treats both simultaneously, continuing to adjust colours well past generation 1 000 (20).

SSIM performs worst across all metrics, nearly double the error of the other two GAs on RMSE and CIEDE2000, with a negligible SSIM advantage (0.532 vs ≈ 0.51). The fundamental issue is what SSIM penalises: it rewards matching local structural patterns and contrast but does not penalise absolute colour error directly. The GA learns structure without correcting colour: a recognisable silhouette from generation 500 with wrong colours that remain unchanged through generation 3 000 (20). The population dynamics show SSIM loses diversity earlier and more severely than the other two GAs, without the oscillatory recovery visible in CIEDE2000 (22).

5.2 Part II - Multi-Objective Optimisation (NSGA-II)

Since neither RMSE nor CIEDE2000 dominates the other, the natural extension is to optimise both simultaneously. NSGA-II [7] ranks individuals by Pareto dominance and uses crowding distance to maintain diversity along the front; all other operators remain unchanged. SSIM was excluded as a third objective given its inconsistent signal.

The Pareto front maps solutions where no improvement in pixel accuracy is possible without sacrificing perceptual colour and vice versa. Placing the single-objective results in the same space confirms that the two objectives are in genuine tension: the NSGA-II fills the intermediate region that no single-objective approach can produce (25). Its position sits closer to the CIEDE2000 end, consistent with CIEDE2000 providing the stronger optimisation signal when both objectives compete. The NSGA-II best member shows structural details absent from the single-objective CIEDE GA (the white collar and face-background transition are more defined) suggesting the simultaneous RMSE pressure acts as a regulariser on geometry even for solutions that prioritise colour (24).

5.3 Conclusions

The fitness function shapes what the GA learns, not merely how efficiently it learns. CIEDE2000 directs effort towards perceptually salient regions and locks in correct colours earlier, making it the more principled choice when visual resemblance is the goal. SSIM, while theoretically motivated, does not penalise colour error directly and produces structurally plausible but chromatically incorrect results. The NSGA-II makes the RMSE-CIEDE2000 trade-off explicit and navigable, revealing that the two objectives are genuinely conflicting and that each single-objective GA finds an extreme that the other cannot reach.

6 Additional Insights

6.1 Spatial Error Distribution

To understand *where* the GA struggles rather than just how much error remains, we computed per-pixel RMSE heatmaps at selected checkpoints. The results reveal a two-phase convergence

pattern: RMSE drops sharply up to generation 500 as triangles reach plausible positions and colors, then slows as residual error concentrates in fine-detail regions (hair, facial contours, edges) that 100 triangles cannot faithfully reproduce (D.1)

6.2 Draw Order Significance

Since triangles are rendered in sequence (index 0 at the bottom, 99 at the top), the draw order determines which colors are visible in overlapping regions. To test whether the GA learned a meaningful draw order, we generated and compared 200 random permutations of the best individual's triangle list. The original RMSE sits far below all 200 shuffled versions (mean 46.86, std 4.55, $p \approx 10^{-10}$), confirming that the learned ordering carries significant information and the GA exploits occlusion by sequencing background triangles first and detail triangles last (D.2).

6.3 Color Palette Convergence

Tracking the 8 dominant colors (via K-Means clustering in RGB space) across checkpoints shows that palette convergence happens very early. At generation 0 the palette is essentially random; by generation 100 warm browns and muted greens already dominate. From generation 500 onwards the palette stabilises into earth tones closely matching the original painting, with subsequent generations producing only incremental refinement (D.3).

6.4 Generalisation to Other Images

To test whether the best configuration generalises beyond the original target, we applied it unchanged to two additional paintings: the *Mona Lisa* (portrait) and a geometric composition by the Portuguese artist Nadir Afonso (flat colors, hard edges). The GA achieved an RMSE of 19.35 on the *Mona Lisa*, lower than on the original target, confirming good transfer to similar portraits (31). On the Nadir Afonso painting, RMSE was roughly $2.5\times$ higher: a counterintuitive result, since the image appears simpler, but sharp boundaries and non-triangular shapes (circles, rectangles) are hard to approximate with overlapping semi-transparent triangles (32).

7 Conclusions

This project showed that a well-tuned GA can reconstruct a recognisable approximation of Vermeer's painting using only 100 semi-transparent triangles. Three key findings stand out. First, the mutation strategy matters more than any other operator: sigma decay alone accounted for the largest single-phase improvement, confirming that balancing exploration and exploitation over time is critical in high-dimensional continuous spaces. Second, the choice of fitness function fundamentally shapes what the GA learns - CIEDE2000 produces more perceptually faithful colours at the cost of higher pixel error and the NSGA-II front showed this trade-off is genuine and unavoidable with a single objective. Third, the GA learns more than just colours and positions: the draw order shuffle test proved that the algorithm implicitly discovers a meaningful layering strategy, exploiting occlusion as an additional degree of freedom. The main limitations remain the fixed 100-triangle budget, which caps representational detail and the OFAT methodology, which cannot fully capture operator interactions. With more time, increasing the triangle count and running more seeds per configuration would strengthen both the results and the analysis.

References

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A System Design and Implementation

A.1 Representation

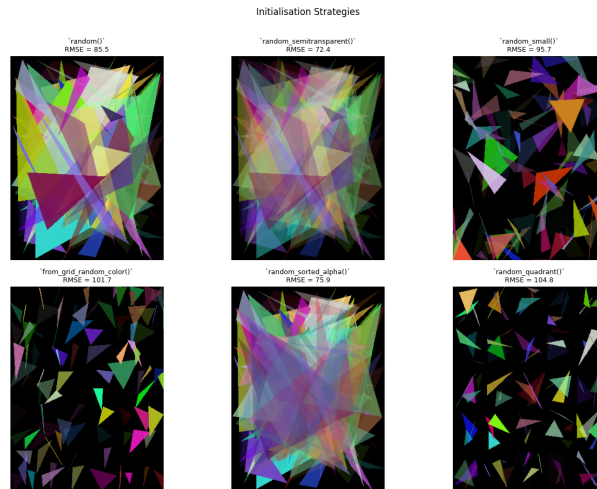


Figure 1: Initialisation Strategies

A.2 Rendering Pipeline



Figure 2: Layered Rendering

A.3 Genetic Operators

A.3.1 Selection

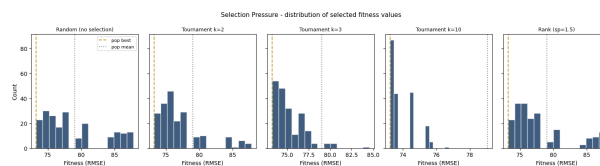


Figure 3: Selection Pressure - Distribution of selected fitness values

A.3.2 Crossover

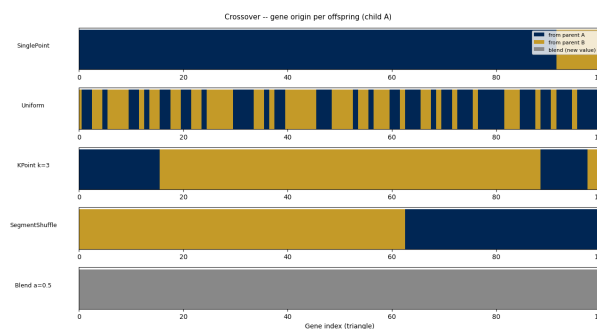


Figure 4: Crossover - Gene origin per offspring

A.3.3 Mutation



Figure 5: Mutation - Rendered result and number of genes modified

B Results and Analysis

B.1 Baseline



Figure 6: Baseline: Original vs GA Approximation

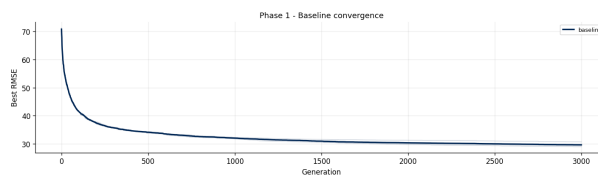


Figure 7: Baseline Convergence

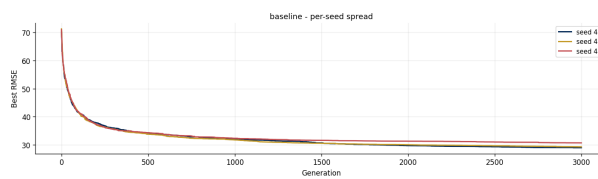


Figure 8: Baseline Per Seed Spread

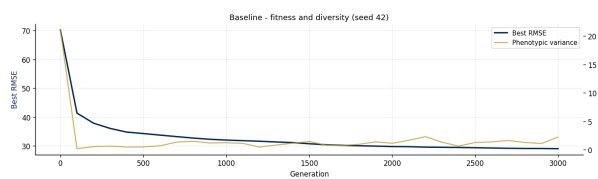


Figure 9: Baseline - Fitness and Diversity (seed 42)

B.2 Mutation Sweep (Phase 5)



Figure 10: Phase 5: Original vs GA Approximation

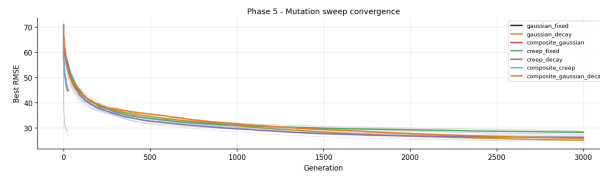


Figure 11: Mutation Sweep Convergence

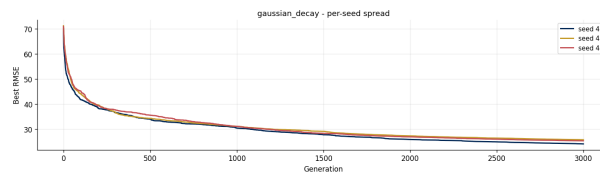


Figure 12: GA with Gaussian Decay - Per Seed Spread

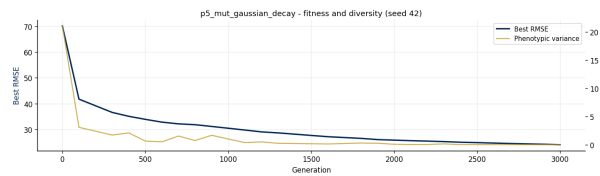


Figure 13: GA with Gaussian Decay - Fitness and Diversity (seed 42)

B.3 Final Results

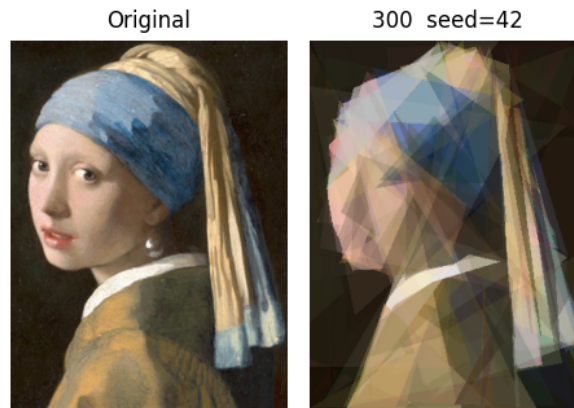


Figure 14: Best Result: Original vs GA Approximation

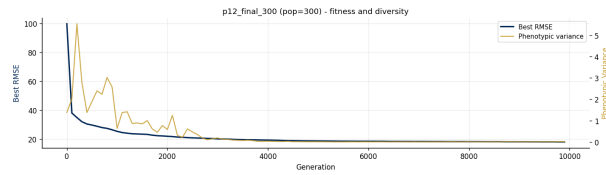


Figure 15: Convergence Curve

B.4 Statistical Analysis

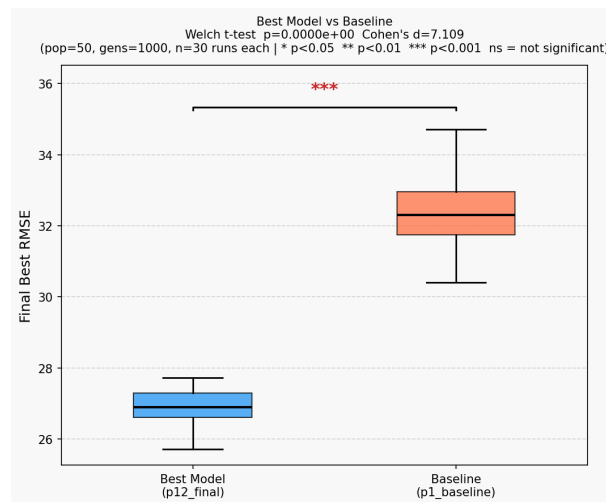


Figure 16: Boxplot: Best Model vs Baseline

C Challenge 1: Alternative Fitness Functions

C.1 Part I - Single-Objective Fitness Comparison

GA	Mean (gens)	Std
RMSE	1736	99
CIEDE2000	1998	17
SSIM	869	172

Figure 17: Convergence speed: mean number of generations to reach 110% of final best value per fitness function.

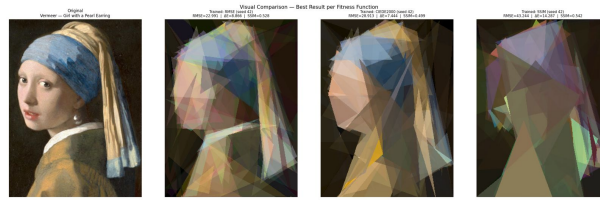


Figure 18: Visual comparison of best individual per fitness function (seed 42).

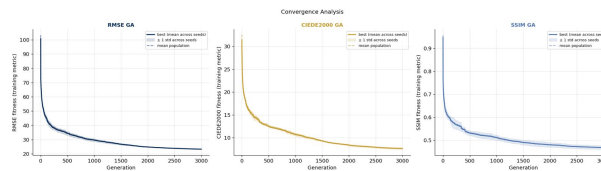


Figure 19: Convergence curves for each fitness function across seeds 42, 43, 44.

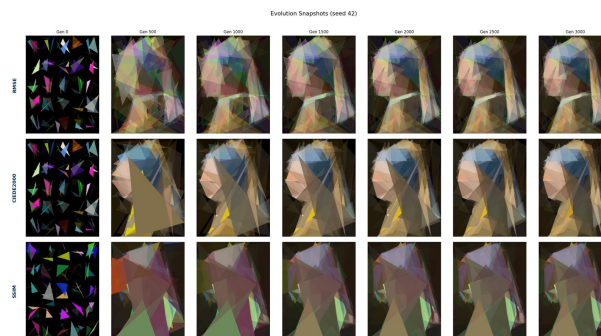


Figure 20: Evolution snapshots at fixed generation intervals for each fitness function (seed 42).

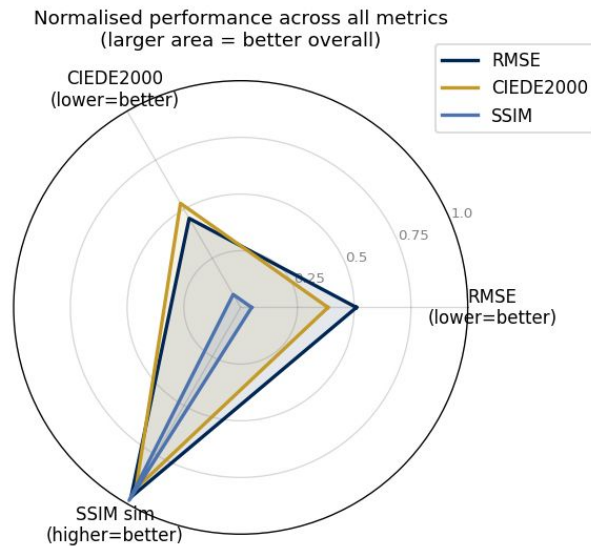


Figure 21: Radar chart comparing RMSE, CIEDE2000 and SSIM GAs across all three evaluation metrics.

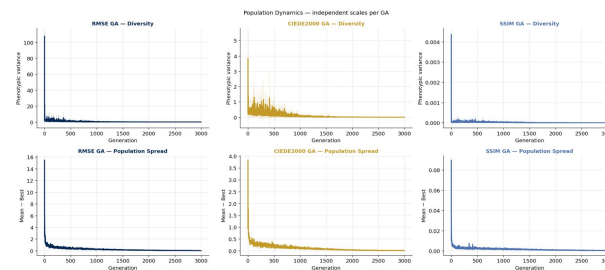


Figure 22: Population dynamics: phenotypic variance and mean–best spread over generations for each fitness function.

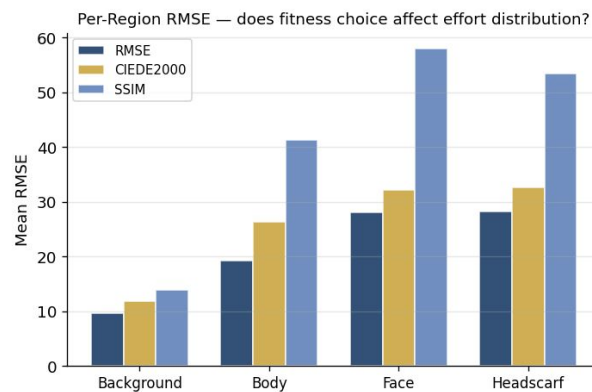


Figure 23: Per-region RMSE analysis across Background, Face, Headscarf and Body for each fitness function.

C.2 Part II - Multi-Objective Optimisation (NSGA-II)



Figure 24: NSGA-II visual results: best CIEDE member compared against single-objective GA results.

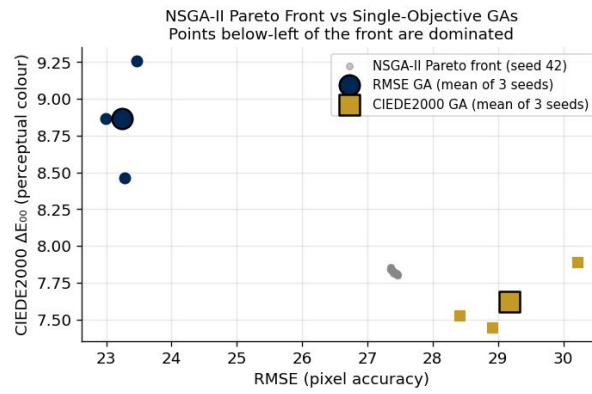


Figure 25: NSGA-II vs single-objective GAs in RMSE×CIEDE2000 objective space.

D Additional Insights

D.1 Spatial Error Distribution

RMSE error heatmap evolution - seed 42 | left: rendered image right: per-pixel error

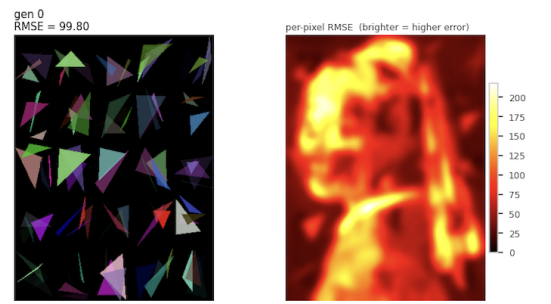


Figure 26: RMSE Error Heatmap Evolution - gen 0

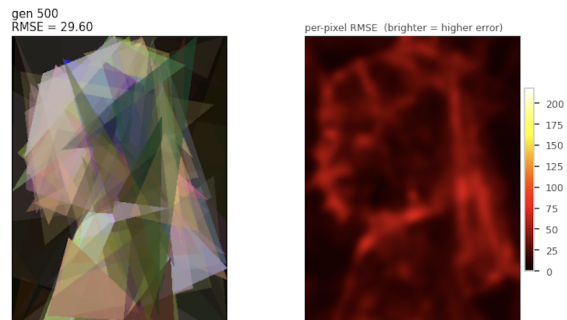


Figure 27: RMSE Error Heatmap Evolution - gen 500

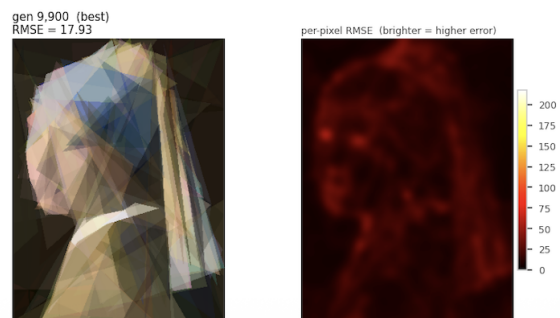


Figure 28: RMSE Error Heatmap Evolution - gen 9000

D.2 Draw Order Significance

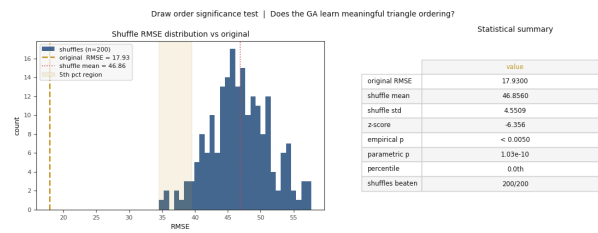


Figure 29: Draw Order Significance Test

D.3 Color Palette Convergence

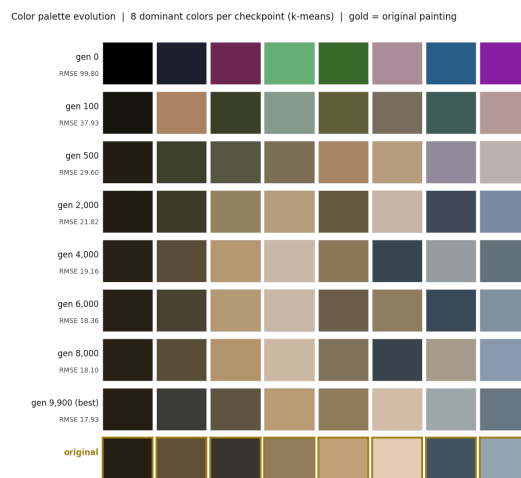


Figure 30: Color Palette Evolution

D.4 Generalisation to Other Images



Figure 31: Mona lisa: Original vs GA Approximation



Figure 32: Nadir Afonso: Original vs GA Approximation